FINAL YEAR PROJECT PROPOSAL

Proposal Title

AI-Based Cotton Leaf Disease Detection and Treatment Recommendation System

Project Brief

Cotton is a significant agricultural product in Pakistan, and there are many diseases which have negative effects on the overall health of cotton plants and the amount of cotton produced. For this reason, we are developing an Android based application that utilizes deep learning and computer vision to identify certain types of cotton leaf conditions based on a photograph. To use the application, a user simply takes a picture of a cotton plant, or chooses an existing picture from their device's gallery. Once the image is processed, the application displays the predicted class of cotton leaf condition along with a level of confidence in the prediction, and provides basic information regarding prevention and control measures for that particular condition. The system is intended to serve as a preliminary decision-making aid, but does not provide diagnostic services, nor does it provide recommendations for treating conditions.

Project Description

The first step is to check over and prepare the labeled cotton leaf image dataset in order to use it to train an image classification algorithm. Some possible categories include "Healthy Cotton Leaf", "Cotton Leaf Curl Virus", "Bacterial Blight", "Alternaria Leaf Spot", and "Cercospora Leaf Spot". However, the final categories will depend upon whether there are enough verified labeled images available. In addition, we may apply a CNN (convolutional neural network) and/or a transfer learning model like EfficientNetB0 to train and evaluate on the test set. Finally, the Android app will allow users to take pictures of leaves or upload existing ones.

Technologies

Python; TensorFlow/Keras; CNN or EfficientNetB0; Kotlin; Android Studio; OpenCV/Pillow; FastAPI (if required); Git and GitHub. Firebase or Room may be considered for optional scan history.

Preliminary Dataset Sources

Kaggle: https://www.kaggle.com/seroshkarim/cotton-leaf-disease-dataset
Zenodo: https://zenodo.org/records/17011709
Mendeley Data: https://data.mendeley.com/datasets/t9hgvk2h9p/1
Dataset class labels, image counts, and licence must be checked before use.